Semiparametric Modelling of Seasonal Time Series

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1. A semiparametric generalization of the Berlin Method

About thirty years ago an approach for modelling seasonal time series, called the **Berlin Method** (BV, Berliner Verfahren) was developed here in Berlin (Heiler, 1970), which in its fourth version is being applied by the German Statistical Office since 1983. In recent years, some data driven algorithms of this approach were developed (Heiler and Feng, 2000, Feng, 1999). In this paper a semiparametric version of the BV is proposed following Beran (1999):

$$Y_t = g(x_t) + S_t^* \cdot g_s(x_t) + X_t, t = 1, 2, ..., n, \tag{1}$$

where $x_t = t/n$, $g(x)$ is a smooth trend, $\{X_t\}$ is a stationary, invertible ARMA(r, s) error process with autocovariances $\gamma(k)$ and $S(x_t, t) := S_t^* \cdot g_s(x_t)$ is a slowly changing seasonal component, where S_t^* is exactly periodic with period T and $g_s(x) > 0$ is a smooth trend in S.

2. Estimation and asymptotic properties

Given observations $y_1, y_2, ..., y_n$. Following the BV, $g(x)$ and $S(x_t, t)$ are estimated based on local regression with polynomials and trigonometric functions as regressors by minimizing

$$Q(x_t) = \sum_{i=1}^{n} \left\{ y_t - \sum_{j=0}^{p} \beta_{1j}(x_i - x_t)^j - \sum_{j=1}^{q} [\beta_{2j} \cos \lambda_j(i-t) + \beta_{3j} \sin \lambda_j(i-t)] \right\}^2 K\left(\frac{x_i - x_t}{h}\right), \tag{2}$$

where $K(u)$ is a second order kernel, h is the bandwidth, $\lambda_1 = 2\pi/T$ and $\lambda_j = j\lambda_1$, for $j = 2, ..., q$, where $q = [T/2]$ and $[\cdot]$ denotes the integer part. The last sin-term has to be omitted, if T is even. The solutions of (2) are $\hat{g}(x_t) = \hat{\beta}_{10}(x_t)$ and $\hat{S}(x_t, t) = \sum_{j=1}^{q} \hat{\beta}_{2j}(x_t)$. Compared with another local regression approach, STL in Cleveland et al. (1990), the BV has some advantages: 1. The variance of $\hat{S}$ is smaller, 2. $\hat{g}$ and $\hat{S}$ are asymptotically independent and 3. The AMISE's (asymptotic mean integrated squared error) of $\hat{g}$ and $\hat{S}$ are summable.

Assume that p is odd. The asymptotic bias and variance of $\hat{g}$ at $0 < x < 1$ are given by

$$E[\hat{g}(x) - g(x)] \doteq \frac{I(K_p) g^{(p+1)}(x)}{(p+1)!} h^{(p+1)} \qquad \text{and} \qquad \text{var}\,[\hat{g}] \doteq \frac{2\pi}{nh} c_f R(K_p), \tag{3}$$

where $K_p(u)$ denotes the equivalent kernel of $\hat{g}$, $R(K_p) = \int K_p^2(u)du$, $I(K_p) = \int u^{p+1} K_p(u)du$ and $c_f = (2\pi)^{-1} \sum_{k=-\infty}^{\infty} \gamma(k)$. The weights of $\hat{S}$ are asymptotically the kernel weights in (2) multiplied by a periodic term (Feng, 1999). At $nh < t < n(1-h)$, this result leads to

$$E[\hat{S}(x_t, t) - S(x_t, t)] \doteq \frac{I(K) g_s''(x_t)}{2} h^2 \qquad \text{and} \qquad \text{var}\,(\hat{S}) \doteq \frac{2\pi}{nh} \left[T c_f^* - c_f \right] R(K), \tag{4}$$

where $R(K)$ and $I(K)$ are defined similarly as in (3), and $c_f^* = (2\pi)^{-1} \sum_{k=-\infty}^{\infty} \gamma(kT)$. The AMISE's of $\hat{g}$ and $\hat{S}$ on $[\Delta, 1-\Delta]$ can be calculated from (3) and (4), where $\Delta > 0$ is a small number. Let $m(x_t, t) = g(x_t) + S(x_t, t)$. The AMISE of $\hat{m}$ is the sum of those of $\hat{g}$ and $\hat{S}$.

Following the idea of Beran (1999), we propose to estimate the unknown ARMA parameter vector $\theta = (\phi_1, ..., \phi_r, \psi_1, ..., \psi_s)'$ from the residuals $r_t = y_t - \hat{m}(x_t, t)$ by maximum likelihood. For $\hat{\theta}$ we have: 1. $\text{var}\,(\hat{\theta}) = O(n^{-1})$, if $\hat{m}$ is consistent, 2. $E(\hat{\theta} - \theta) = \max\{O[(nh)^{-1}], h^2\}$ due to the bias and variance in $\hat{g}$ and $\hat{S}$, and 3. $\hat{\theta}$ is $\sqrt{n}$-consistent, if $O(n^{-1/2}) \le h \le O(n^{-1/4})$. Note that $\hat{\theta}$ obtained following the algorithms described below is not $\sqrt{n}$-consistent.

3. Data driven algorithms and application

Three iterative data-driven algorithms are developed for practical implementation, where the AMISE's of $\hat{g}$, $\hat{S}$ as well as $\hat{m}$ are taken to be the optimal criteria, respectively. A proper procedure was developed for estimating $\hat{g}_s''$. In each iteration, c_f and c_f^* are estimated nonparametrically from r_t, so that the algorithms are independent of the parametric assumption and are hence more stable. The parameter vector θ is estimated at the end of the procedure.

The proposal is applied to the time series of the non-seasonally adjusted, monthly global land temperature from Jan. 1880 to Dec. 2002 (source: the National Climatic Data Center). Figure 1 shows the local linear $\hat{g}$ together with the seasonally adjusted data (left) and the ACF of the residuals (right), where the bandwidth was selected using the AMISE of $\hat{g}$. The data and the estimated seasonal component are omitted. Following the BV, $\hat{g}$ and $\hat{S}$ are both estimated directly from the data. The seasonal adjustment and the trend elimination for estimating g rep. S are carried out automatically. An ARMA(1, 1) model $X_t = 0.6978X_{t-1} - 0.3395\epsilon_{t-1} + \epsilon_t$ was fitted to the residuals, where the orders were selected from $r, s = 0, 1, ..., 5$, following the BIC. This example shows that the proposal works well in practice.

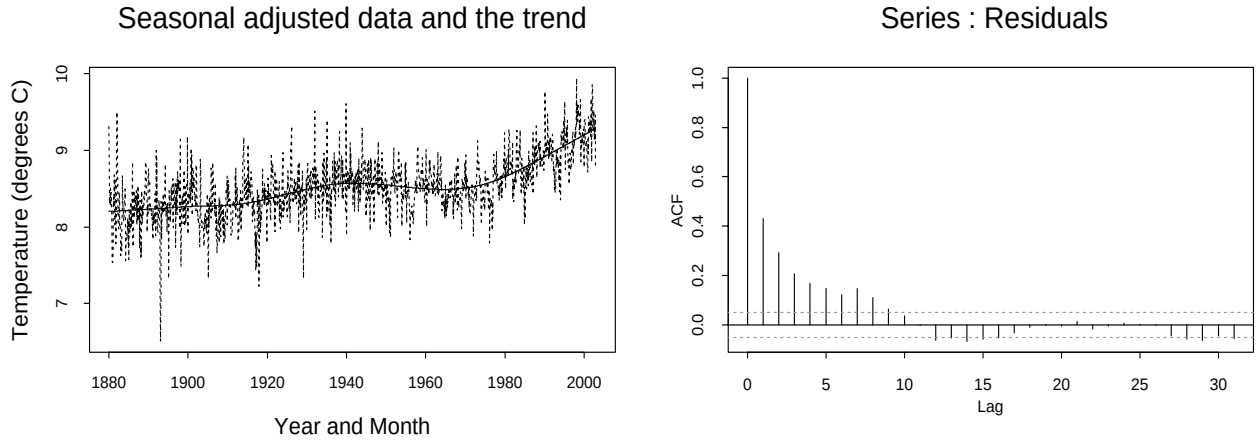


Figure 1. $\hat{g}$ and the seasonally adjusted data (left), and the ACF of r_t (right).

REFERENCES

Beran, J. (1999). SEMIFAR models - A semiparametric framework for modelling trends, long range dependence and nonstationarity. Discussion paper, CoFE, 99/16, Univ. of Konstanz.

Cleveland, R.B., Cleveland, W.S., McRae, I.E. and Terpenning, I. (1990). STL: A seasonal-trend decomposition procedure based on LOWESS. *J. Official Statistics*, **6**, 3–73.

Feng, Y. (1999). Kernel- and Locally Weighted Regression – with Application to Time Series Decomposition. Verlag für Wissenschaft und Forschung, Berlin.

Heiler, S. (1970). Theoretische Grundlagen des "Berliner Verfahrens". In Sonderheft 1 zum Allg. Statist. Archiv (edited by W. Wetzel), 67–93.

Heiler, S. and Feng, Y. (2000). Data-driven decomposition of seasonal time series. J. Statist. Plann. Infer., 91, 351 – 363.

RÉSUMÉ

Cet article propose une extension semi-paramétrique de la méthode de Berlin concernant la décomposition des séries chronologiques saisonnières. L'estimation du modéle ainsi que des propriétés asymptotiques sont étudiées en détail. Nous développons également des algorithmes dirigés par les donneés et illustrons cette approche par l'étude d'un exemple.